
CS 559: CREDIT RISK DEFAULT MODELING

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Abstract

This project presents a comprehensive framework for **Credit Risk Modeling** that integrates data preprocessing, probability estimation, and loss quantification. Using a large-scale loan-level dataset derived from LendingClub-style records, we develop a streamlined and leakage-safe dataset (`credit_risk_master_50k.csv`) engineered to support modeling of **Probability of Default (PD)**, **Exposure at Default (EAD)**, and **Loss Given Default (LGD)**. The pipeline involves rigorous data cleaning, imputation, correlation-based feature pruning, and redundancy elimination to ensure both accuracy and regulatory transparency.

We implement a dual-stage modeling approach: a *baseline logistic regression* for interpretable PD estimation, and a *gradient boosting ensemble* (LightGBM/XGBoost) for optimized predictive performance. Future extensions incorporate EAD and LGD estimation using regression-based techniques, culminating in the computation of *Expected Loss* ($EL = PD \times LGD \times EAD$) as the principal credit risk measure.

1 Introduction

Credit risk quantifies both the likelihood and severity of borrower default. Banks and regulators decompose risk via three parameters: **PD** (probability that a borrower defaults), **EAD** (exposure outstanding at default), and **LGD** (loss rate conditional on default). These combine into the *Expected Loss*:

$$EL = PD \times LGD \times EAD. \quad (1)$$

Our goal is to develop an end-to-end **Credit Risk Modeling framework** that accurately quantifies borrower default risk and potential financial losses. This involves constructing a robust, well-structured dataset and implementing predictive models to estimate the **Probability of Default (PD)**, while retaining all necessary fields to derive **Exposure at Default (EAD)** and **Loss Given Default (LGD)**. We emphasize data integrity, model interpretability, and statistical soundness to ensure that each stage, from preprocessing and feature selection to PD estimation, supports reliable, transparent, and regulator-ready credit risk evaluation.

2 Related Work

Prior studies use public loan-level data (e.g., LendingClub-style datasets) to build PD classifiers (logistic regression, tree ensembles, gradient boosting) and regression-style EAD/LGD estimators. Common pitfalls include including post-outcome variables (*data leakage*) and insufficient feature pruning. Our work addresses these by enforcing strict temporal logic, removing outcome-proximal fields, and applying correlation-based pruning to keep only predictive, non-redundant features.

3 Methodology

This project formulates credit risk assessment as a supervised learning problem centered on predicting the Probability of Default (PD), while preparing the dataset for future computation of Exposure at Default (EAD) and Loss Given Default (LGD). The methodological design ensures both predictive accuracy and regulatory interpretability.

3.1 Problem Formulation

Each loan record is represented as a feature vector $\mathbf{x}_i \in \mathbb{R}^d$ describing borrower and loan characteristics (e.g., interest rate, income, credit utilization, grade, and term). The target variable $y_i \in \{0, 1\}$ denotes whether a borrower defaulted:

$$y_i = \begin{cases} 1, & \text{if the loan is charged off or defaulted,} \\ 0, & \text{if the loan is fully paid or current.} \end{cases}$$

The objective is to learn a function

$$\hat{p}_i = f(\mathbf{x}_i; \theta)$$

that outputs the estimated probability of default $\hat{p}_i = P(y_i = 1|\mathbf{x}_i)$ for each borrower.

3.2 Data Processing and Feature Engineering

Starting from the raw $999,999 \times 145$ LendingClub-style dataset, we executed the following pipeline:

1. **Data Cleaning:** Removed exact duplicates and non-informative identifiers (`id`, `member_id`, `url`). Percentages in numeric columns (e.g., `int_rate`, `revol_util`) were standardized to decimals, and date fields were parsed.
2. **Missing Value Handling:** Numeric attributes were imputed with median values; categorical attributes used an “Unknown” label. Columns with more than 90% missing data (e.g., `hardship` and secondary-applicant fields) were discarded.
3. **Leakage Prevention:** Any variable containing post-outcome or repayment information was excluded to ensure temporal integrity.
4. **Correlation-Based Pruning:** We computed the Pearson correlation matrix among numeric predictors and eliminated highly correlated features ($|r| > 0.85$) to prevent redundancy and multicollinearity.
5. **Feature Consolidation:** The remaining 30 core predictors were reduced to 22 key features after pruning. These capture credit profile, loan terms, and repayment capacity.

3.3 Modeling Approach

The PD estimation problem was approached in two stages:

Baseline Model — Logistic Regression. A regularized logistic regression model served as the benchmark, producing interpretable coefficients and well-calibrated probability estimates:

$$P(y_i = 1|\mathbf{x}_i) = \frac{1}{1 + e^{-(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta})}}.$$

Coefficients $\boldsymbol{\beta}$ quantify marginal effects of features on default probability.

Optimal Model — Gradient Boosting (LightGBM/XGBoost). To capture nonlinear relationships and feature interactions, we employed a gradient boosting ensemble with balanced class weights and early stopping. This approach optimizes the binary cross-entropy loss:

$$\mathcal{L} = - \sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)].$$

The model outputs well-discriminated PDs, later calibrated using isotonic regression to ensure that predicted probabilities match observed default frequencies.

3.4 Preparation for EAD and LGD Estimation

Although PD modeling is the immediate focus, the dataset retains the necessary pre-default financial variables to compute:

$$EAD_i = \text{funded_amnt}_i - \text{total_rec_prncp}_i, \quad LGD_i = 1 - \frac{\text{total_rec_prncp}_i + \text{recoveries}_i}{\text{funded_amnt}_i}.$$

These definitions enable future modeling of expected losses:

$$EL_i = PD_i \times LGD_i \times EAD_i.$$

3.5 Outcome

The final processed dataset (`credit_risk_master_50k.csv`) includes 50,000 representative observations and a compact, leakage-free feature set ready for PD model training, EAD/LGD derivation, and subsequent stress-testing analyses.

4 Experimental Setup

4.1 Data

Starting from the original $999,999 \times 145$ LendingClub-style dataset, we constructed a clean, representative dataset of 50,000 records, titled `credit_risk_master_50k.csv`. The final dataset satisfies the following criteria:

- **No missing values:** All numeric features were imputed via median values; categorical features were assigned an “Unknown” label.
- **No duplicate records:** All exact and ID-based duplicates were removed.
- **Leakage-safe features:** All variables containing post-outcome or repayment information were excluded to ensure model validity.
- **Correlation-pruned structure:** Redundant and collinear features ($|r| > 0.85$) were eliminated to improve generalization and prevent overfitting.

The dataset retains a curated set of columns necessary for both present and future modeling stages:

- **PD modeling:** `term`, `int_rate`, `grade`, `sub_grade`, `purpose`, `emp_length`, `home_ownership`, `annual_inc`, `verification_status`, `dti`, `revol_util`, `pub_rec`, `delinq_2yrs`, `inq_last_6mths`, `revol_bal`, `all_util`, `total_bc_limit`, `addr_state`, `zip_code`, `mths_since_last_record`, and the target `default_flag`.
- **EAD/LGD estimation (future):** `funded_amnt`, `total_rec_prncp`, and `recoveries`.

This unified dataset structure allows end-to-end computation of default probabilities, loss components, and eventual expected loss:

$$EL = PD \times LGD \times EAD.$$

4.2 Evaluation Metrics

For PD modeling, performance will be evaluated using:

- **Accuracy and F1-score** — to assess classification quality.
- **ROC-AUC and KS statistic** — to evaluate the model's ability to separate default and non-default borrowers.
- **Calibration plots** — to ensure predicted probabilities reflect true observed default rates.

For future EAD/LGD regression models, we will use:

- **Mean Absolute Error (MAE)** and **Root Mean Squared Error (RMSE)** for magnitude accuracy.
- **Stress calibration diagnostics** under simulated macroeconomic downturns to validate portfolio robustness.

4.3 Comparison Methods

We plan to evaluate a sequence of increasingly sophisticated models:

- **Baseline:** Logistic Regression — interpretable and regulator-friendly PD model.
- **Intermediate:** Random Forest — captures nonlinearities with feature importance transparency.
- **Optimal:** Gradient Boosting Machines (LightGBM/XGBoost) — provides superior discrimination and scalability, later calibrated using isotonic regression for probability accuracy.

- **Future Extensions:** Linear and gradient boosting regressors for EAD and LGD estimation, combined with a constant-loss baseline for Expected Loss benchmarking.

Overall, the experimental setup emphasizes both statistical rigor and operational realism, ensuring the final models are transparent, interpretable, and suitable for regulatory-grade credit risk forecasting.

5 Results — Currently

The final dataset is compact, interpretable, and leak-safe. Correlation-based pruning removed redundant features while preserving signal; leakage checks eliminated outcome-proximal columns. The deliverable `credit_risk_master_50k.csv` is thus ready for PD modeling today and for computing (in later phases):

$$EAD = \text{funded_amnt} - \text{total_rec_prncp}, \quad (2)$$

$$LGD = 1 - \frac{\text{total_rec_prncp} + \text{recoveries}}{\text{funded_amnt}}, \quad (3)$$

$$EL = PD \times LGD \times EAD. \quad (4)$$